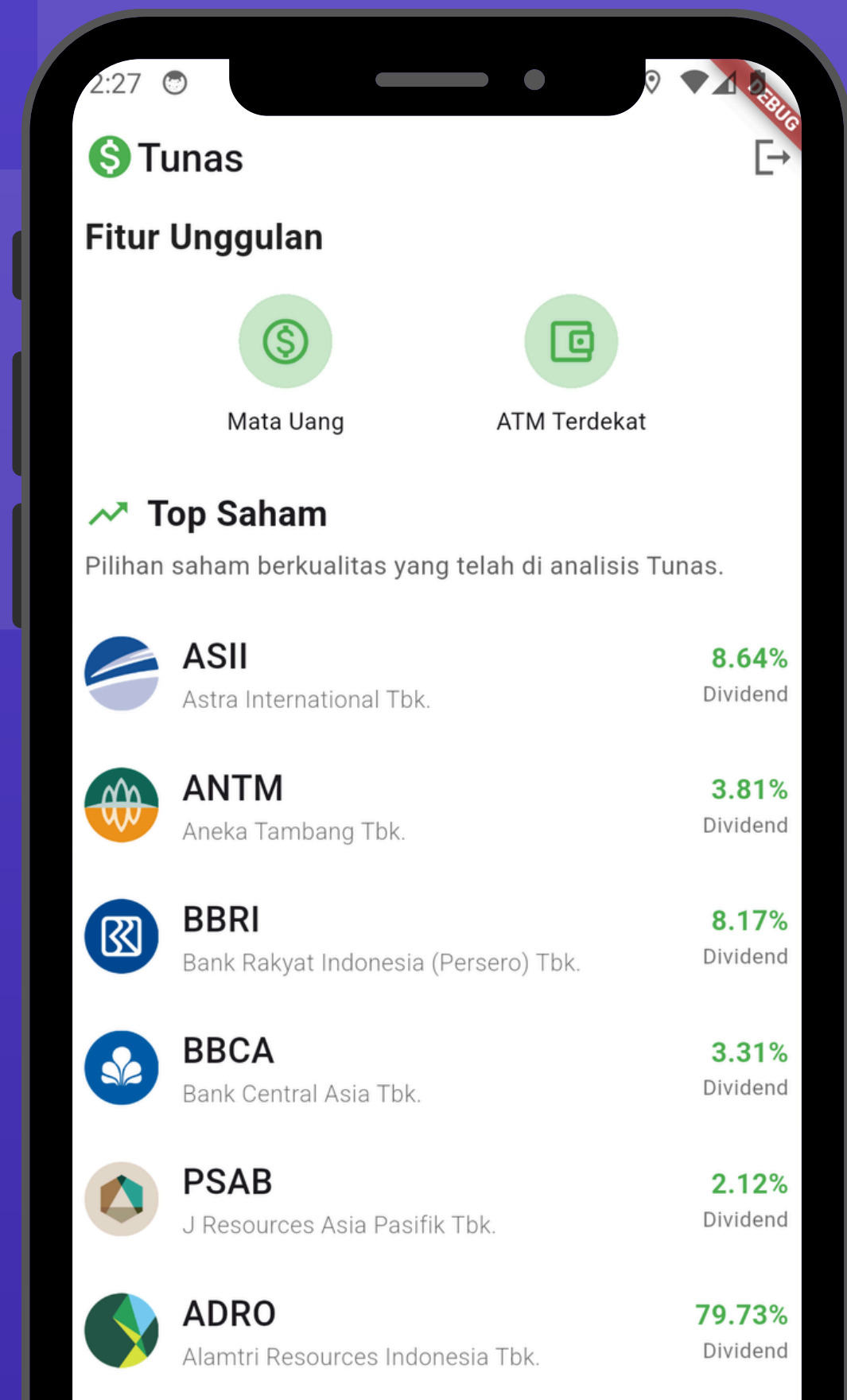


RUPIAH MELEMAH?

Investasi Saham Jadi **Solusi**



IDX Channel > Market News

Jumlah Investor Pasar Modal Indonesia Tembus 20,36 Juta Orang di Akhir 2025

Market News |  Anggie Ariesta | 09/01/2026 10:58 WIB

Pasar modal Indonesia resmi memasuki babak baru dengan mencatatkan basis investor terbesar sepanjang sejarah.



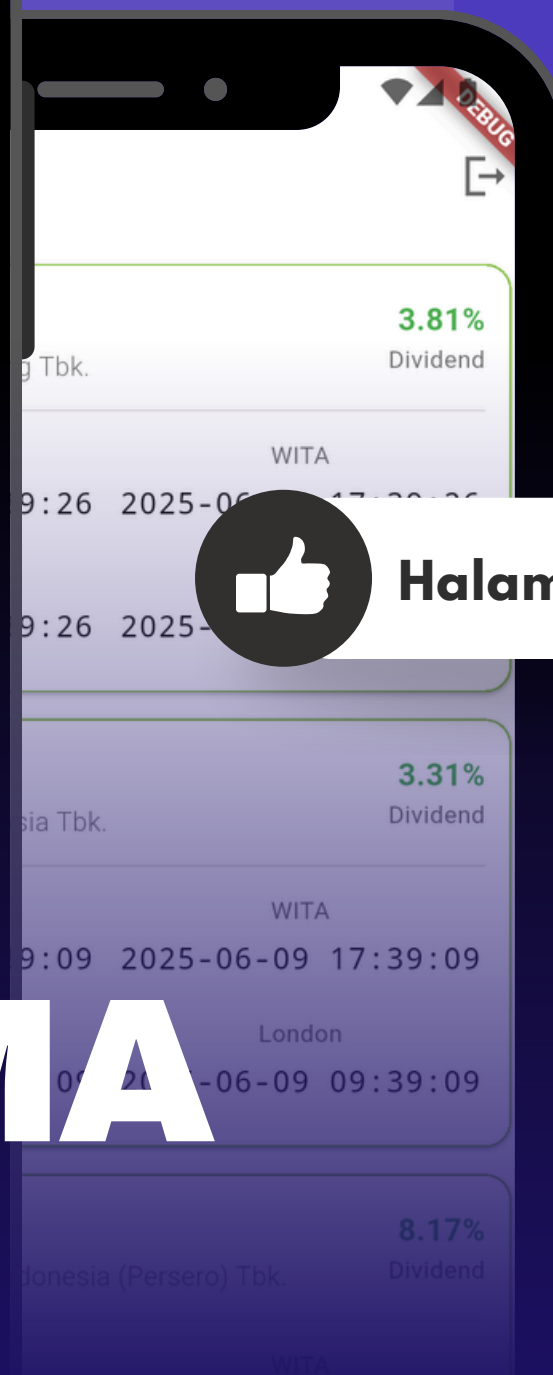
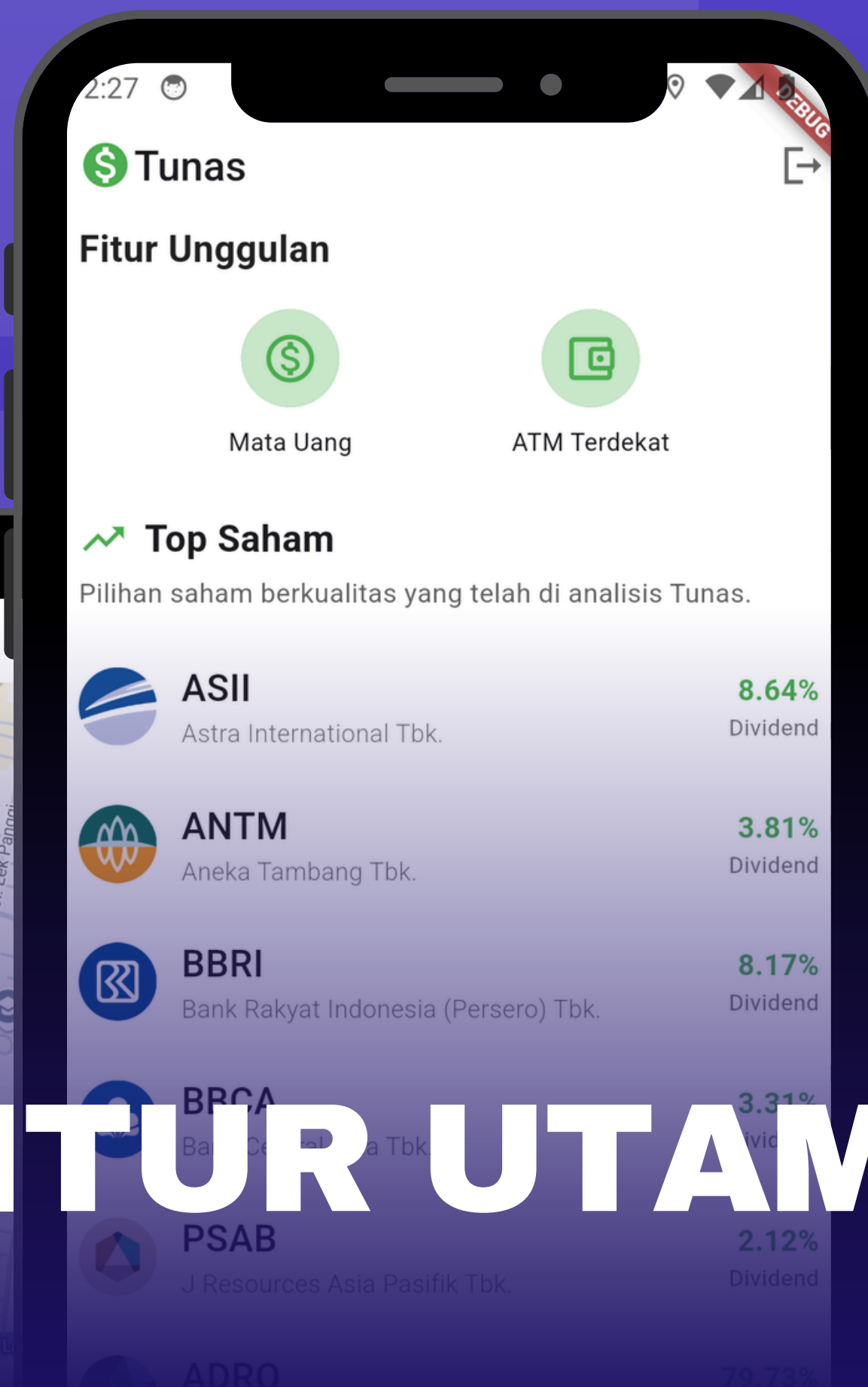
Kompas: Per Agustus 2023, investor individu Gen-Z mencapai 57,04 persen dari total 11,54 juta investor dengan total aset Rp 50,51 triliun..



Register dan Login
Dienskripsi dengan
SHA256

Homepage dengan data
dari **API**

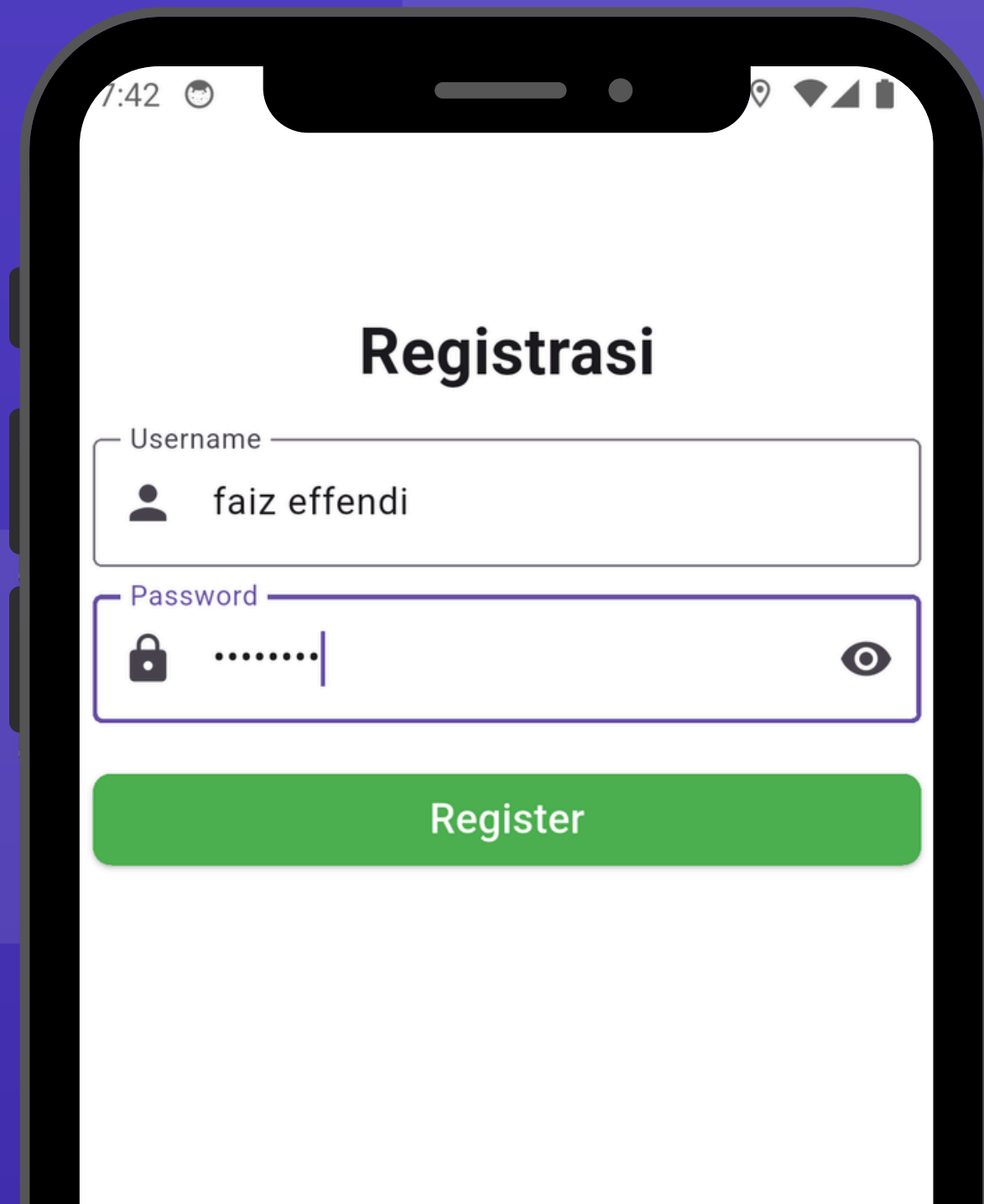
LBS dengan **Google Maps**
menuju **ATM Terdekat**



Halaman Saham **Favorit**

FITUR UTAMA

#1 REGISTER DAN LOGIN



The image shows a smartphone screen with a registration form titled "Registrasi". The form has two input fields: "Username" with the text "faiz effendi" and a person icon, and "Password" with a lock icon and a toggle eye icon. Below the fields is a green "Register" button. The status bar at the top shows the time 7:42 and various icons.

Implementasi Enkripsi dan Session

```
Future<String> _hashPassword(String password) async {  
  var bytes = utf8.encode(password);  
  var digest = sha256.convert(bytes);  
  return digest.toString();  
}  
  
Future<void> _register() async {  
  if (formKey.currentState!.validate()) {  
    SharedPreferences prefs = await SharedPreferences.getInstance();  
    String newUsername = _username.text.trim();  
    String newPassword = _password.text.trim();  
  
    if (prefs.containsKey('password_$newUsername')) {  
      if (!mounted) return;  
      ScaffoldMessenger.of(context).showSnackBar(  
        const SnackBar(  
          content: Text(  
            'Username sudah terdaftar! Silakan gunakan username lain.',  
          ), // Text  
          backgroundColor: Colors.orange,  
        ), // SnackBar  
      );  
      return;  
    }  
  
    String hashedPassword = await _hashPassword(newPassword);  
  
    await prefs.setString('password_$newUsername', hashedPassword);  
    if (!mounted) return;  
    ScaffoldMessenger.of(context).showSnackBar(  
      const SnackBar(  
        content: Text('Registrasi berhasil! Silakan login.'),  
        backgroundColor: Colors.green,  
      ), // SnackBar  
    );  
    Navigator.pop(  
      context,  
    );  
  }  
}
```

#1 REGISTER DAN LOGIN

Signup'. The top status bar shows the time 11:39 and a 'DEBUG' label."/>

11:39

Silahkan Login

Username

Password

Login

Don't have an account? [Signup](#)

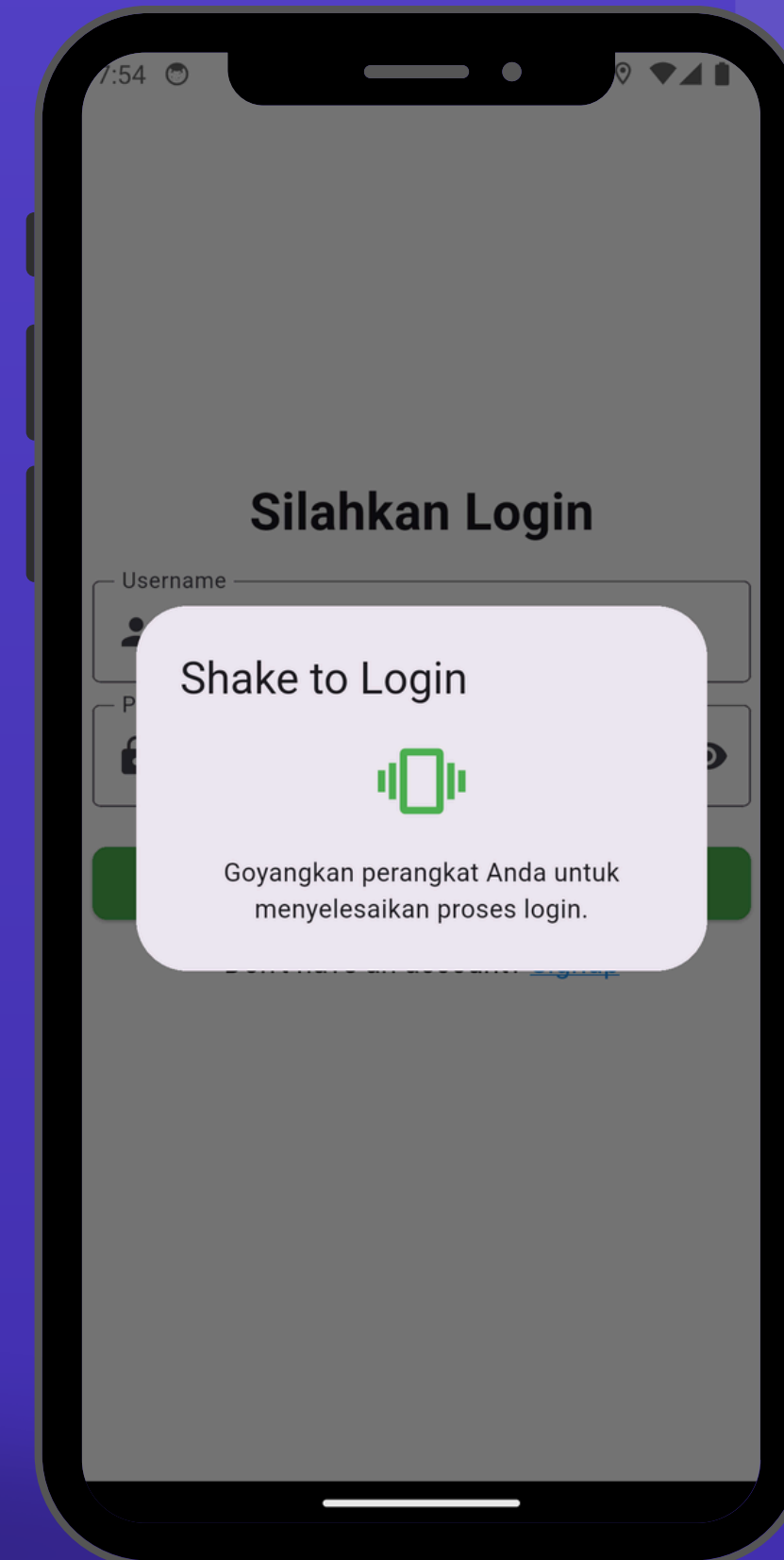
Implementasi Enkripsi dan Session

```
Future<void> _login() async {  
  if (formKey.currentState!.validate()) {  
    SharedPreferences loginData = await SharedPreferences.getInstance();  
    String inputUsername = _username.text.trim();  
    String inputPassword = _password.text.trim();  
  
    String hashedInputPassword = await _hashPassword(inputPassword);  
    String? storedHashedPassword = loginData.getString(  
      'password_$inputUsername',  
    );  
  
    if (storedHashedPassword != null &&  
        storedHashedPassword == hashedInputPassword) {  
      // Simpan username yang sedang aktif untuk digunakan di ProfilePage dll  
      await loginData.setString('username', inputUsername);  
      if (!mounted) return;  
      await Hive.openBox<FavoriteStock>('favorites_$inputUsername');  
      _showShakeDialog();  
    } else {  
      // Login gagal  
      if (!mounted) return;  
      ScaffoldMessenger.of(context).showSnackBar(  
        const SnackBar(  
          content: Text('Login gagal: Username atau password salah'),  
          backgroundColor: Colors.red,  
        ), // SnackBar  
      );  
    }  
  }  
}
```


PENGUNAAN SENSOR ACCELEROMETER

Implementasi Sensor Sederhana

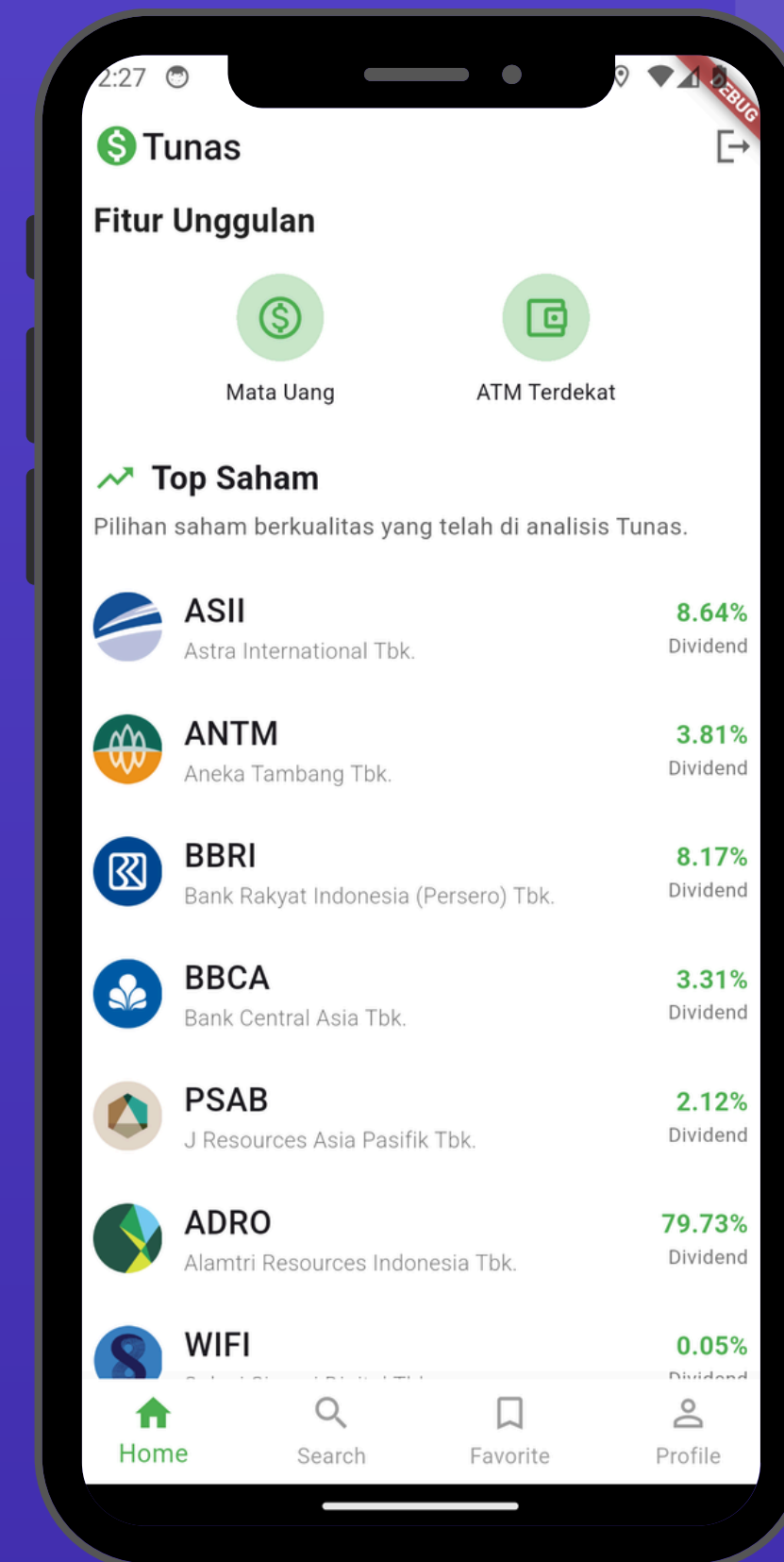
```
builder: (BuildContext context) {  
  ShakeDetector? detector;  
  
  detector = ShakeDetector.autoStart(  
    onPhoneShake: (ShakeEvent event) {  
      detector?.stopListening();  
  
      Navigator.of(context).pop();  
  
      // Show success message  
      ScaffoldMessenger.of(context).showSnackBar(  
        const SnackBar(  
          content: Text('Berhasil login'),  
          backgroundColor: Colors.green,  
        ), // SnackBar  
      );  
  
      // Navigate to the home page  
      Navigator.pushReplacement(  
        context,  
        MaterialPageRoute(builder: (context) => const HomeLayout()),  
      );  
    },  
    shakeThresholdGravity: 1.1,  
  ); // ShakeDetector.autoStart
```



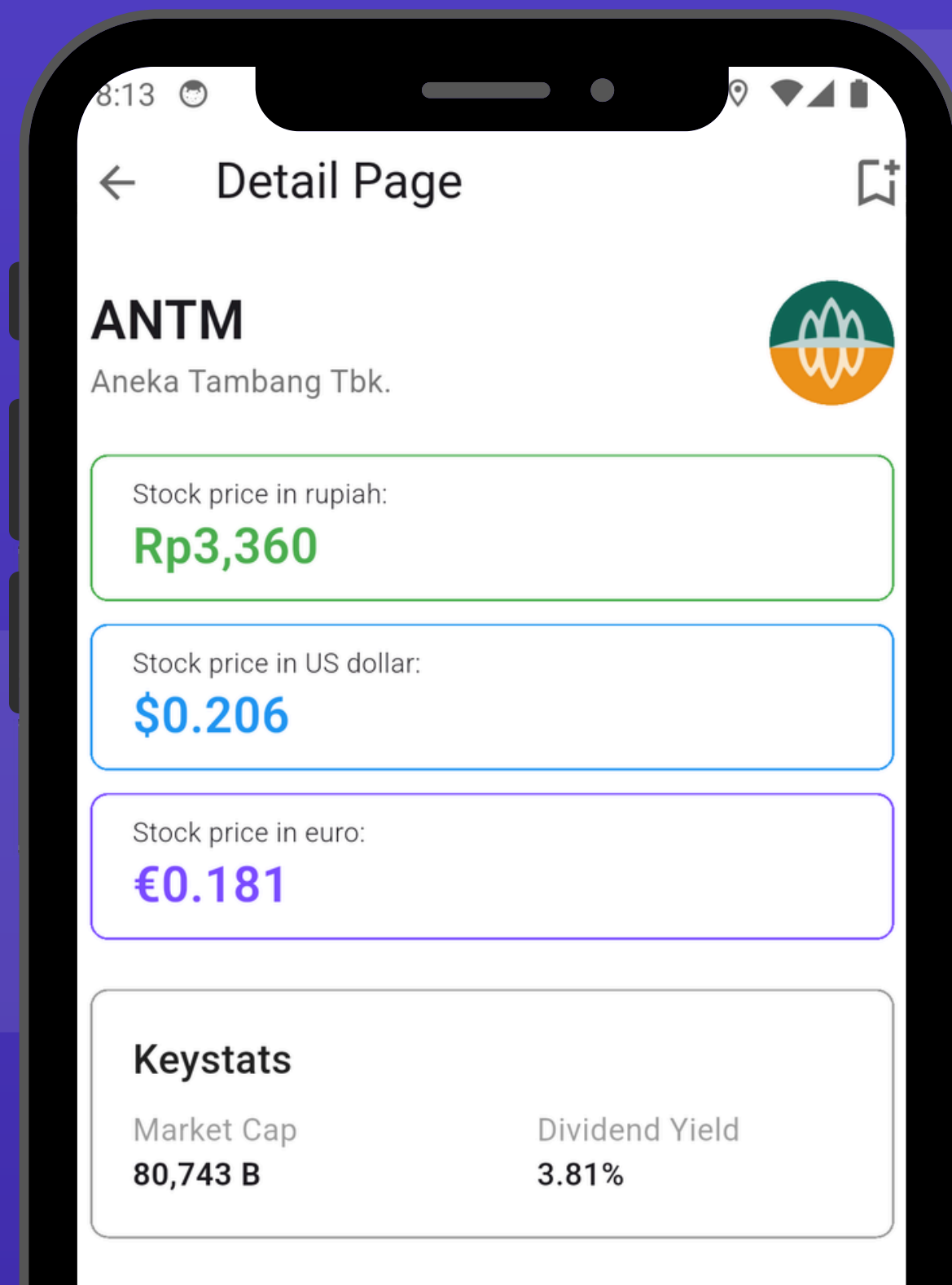
PENGUNAAN API

Menampilkan data dari API

```
FutureBuilder<List<Stock>>(  
  future: stocksFuture,  
  builder: (context, snapshot) {  
    if (snapshot.connectionState == ConnectionState.waiting) {  
      return Center(child: CircularProgressIndicator());  
    }  
    if (snapshot.hasError) {  
      return Center(child: Text("Error: ${snapshot.error}"));  
    }  
    if (snapshot.hasData) {  
      final stocks = snapshot.data!;  
      return ListView.builder(  
        shrinkWrap: true,  
        physics: NeverScrollableScrollPhysics(),  
  
        itemCount: stocks.length,  
        itemBuilder: (context, index) {  
          final stock = stocks[index];  
          return _buildStockItem(context: context, stockData: stock);  
        },  
      ); // ListView.builder  
    }  
    // Default jika tidak ada data  
    return Center(child: Text("Tidak ada data saham."));  
  },  
), // FutureBuilder
```



KONVERSI MATA UANG



Konversi Mata Uang

```
// Convert to Rupiah
Text(
  "\${{(widget.stockData.price / 16300).toStringAsFixed(3)})",
  style: TextStyle(
    fontSize: 22,
    fontWeight: FontWeight.w600,
    color: Colors.blue), // TextStyle
), // Text
```

```
// Convert to Dollar
Text(
  "\${{(widget.stockData.price / 18565).toStringAsFixed(3)})",
  style: TextStyle(
    fontSize: 22,
    fontWeight: FontWeight.w600,
    color: Colors.deepPurpleAccent), // TextStyle
), // Text
```


Hive Database dalam fitur favorit

```
void _checkIfFavorited() {
  if (_favoritesBox != null) {
    _isFavorited = _favoritesBox!.containsKey(widget.stockData.code);
  }
}

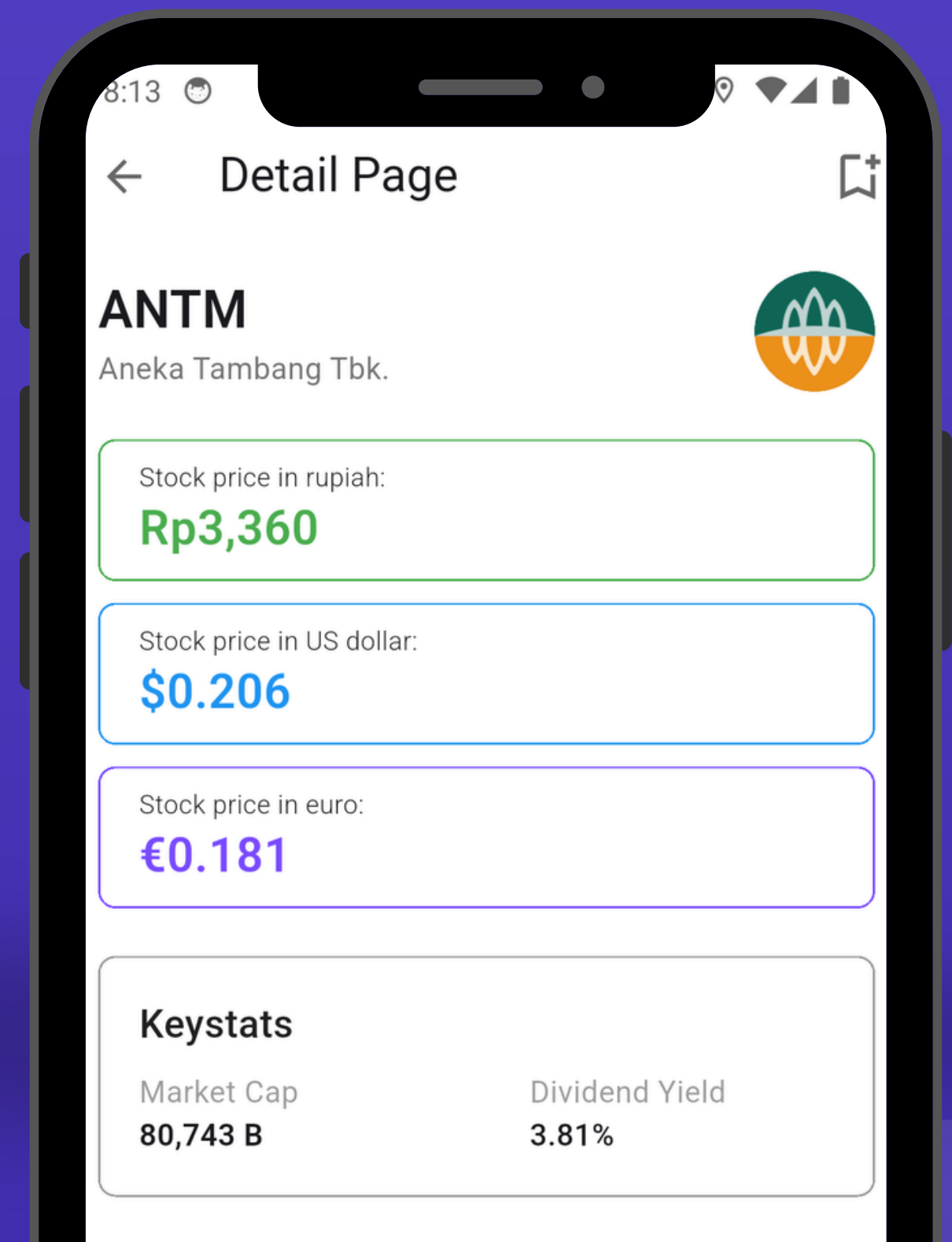
void _toggleFavorite() {
  setState(() {
    _isFavorited = !_isFavorited;

    if (_isFavorited) {
      final nowUtc = DateTime.now().toUtc();
      final formatter = DateFormat('yyyy-MM-dd HH:mm:ss');
      final wibTime = nowUtc.add(const Duration(hours: 7));
      final witaTime = nowUtc.add(const Duration(hours: 8));
      final witTime = nowUtc.add(const Duration(hours: 9));
      final londonTime = nowUtc;

      final newFavorite = FavoriteStock()
        ..id = widget.stockData.id
        ..code = widget.stockData.code
        ..name = widget.stockData.name
        ..imageUrl = widget.stockData.imageUrl
        ..dividend = widget.stockData.dividend
        ..addedAtWIB = formatter.format(wibTime)
        ..addedAtWITA = formatter.format(witaTime)
        ..addedAtWIT = formatter.format(witTime)
        ..addedAtLondon = formatter.format(londonTime);

      _favoritesBox!.put(widget.stockData.code, newFavorite);
    } else {
      _favoritesBox!.delete(widget.stockData.code);
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(
          content: Text('${widget.stockData.code} dihapus dari favorit.'),
          backgroundColor: Colors.redAccent,
        ), // SnackBar
      );
    }
  });
}
```

IMPLEMENTASI HIVE DATABASE, KONVERSI WAKTU, DAN NOTIFIKASI



Menampilkan data dari Hive database

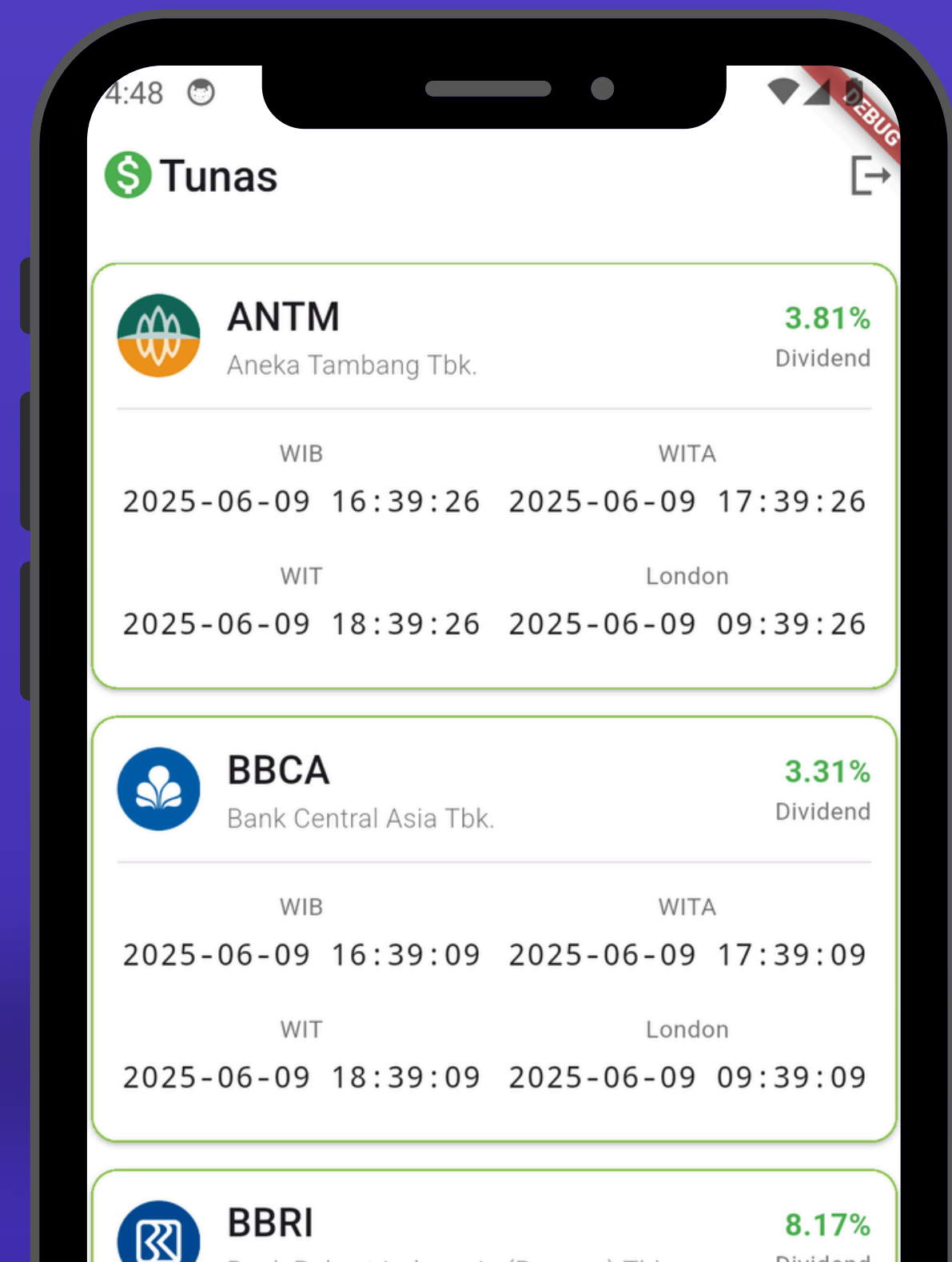
```
@override
Widget build(BuildContext context) {
  if (_favoritesBox == null) {
    return const Center(child: CircularProgressIndicator());
  }

  return ValueListenableBuilder(
    valueListenable: _favoritesBox!.listenable(),
    builder: (context, Box<FavoriteStock> box, _) {
      final favoriteStocks = box.values.toList().cast<FavoriteStock>();

      if (favoriteStocks.isEmpty) {
        return const Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Icon(Icons.bookmark_border, size: 80, color: Colors.grey),
              SizedBox(height: 16),
              Text(
                'Anda belum memiliki saham favorit.',
                style: TextStyle(fontSize: 16, color: Colors.grey),
              ), // Text
            ],
          ), // Column
        ); // Center
      }

      return ListView.builder(
        padding: const EdgeInsets.only(top: 8.0, bottom: 8.0),
        itemCount: favoriteStocks.length,
        itemBuilder: (context, index) {
          final favoriteStock = favoriteStocks[index];
          return _buildFavoriteStockItem(context, favoriteStock);
        },
      ); // ListView.builder
    },
  ); // ValueListenableBuilder
}
```

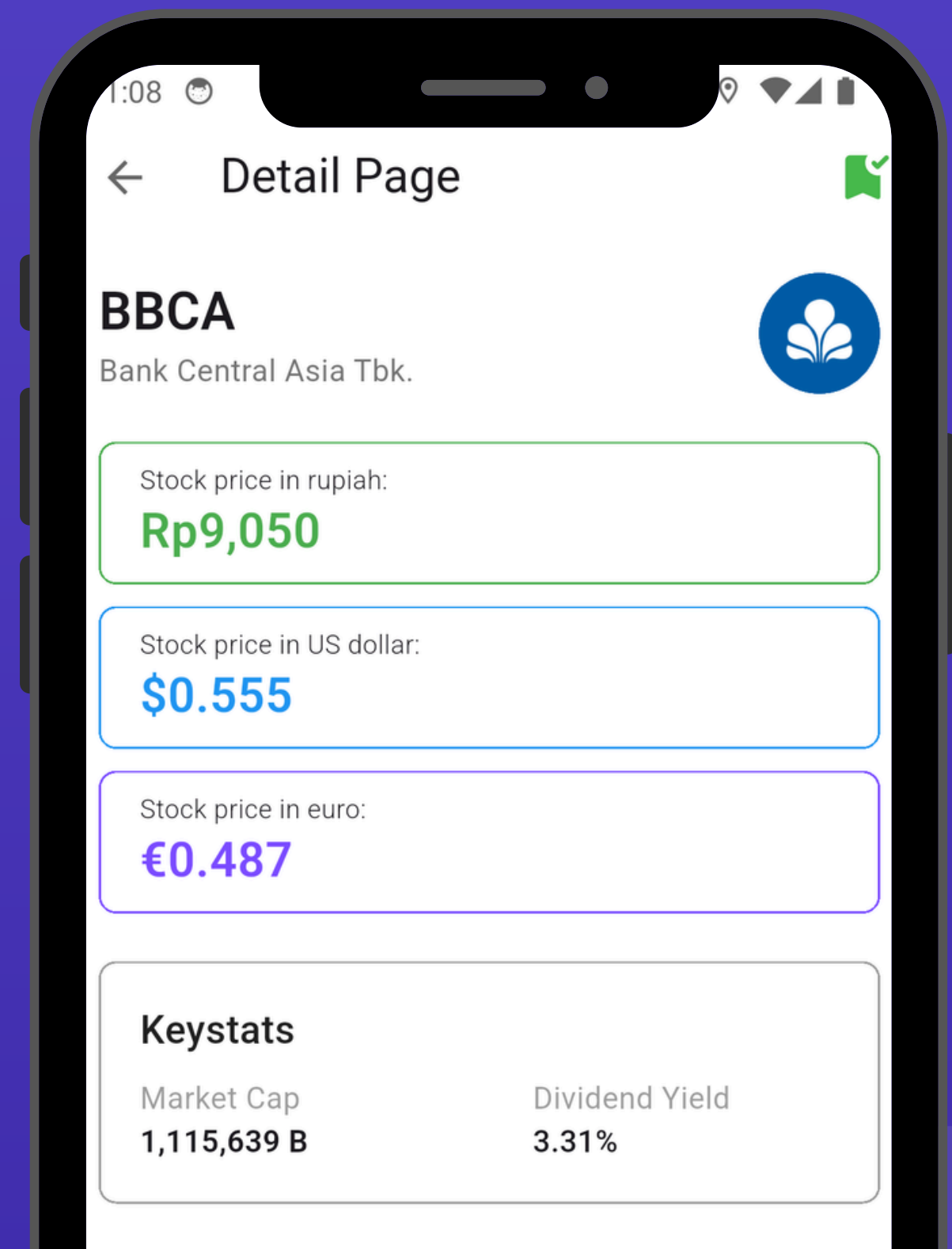
IMPLEMENTASI HIVE DATABASE, KONVERSI WAKTU, DAN NOTIFIKASI



IMPLEMENTASI HIVE DATABASE, KONVERSI WAKTU, DAN NOTIFIKASI

Notifikasi

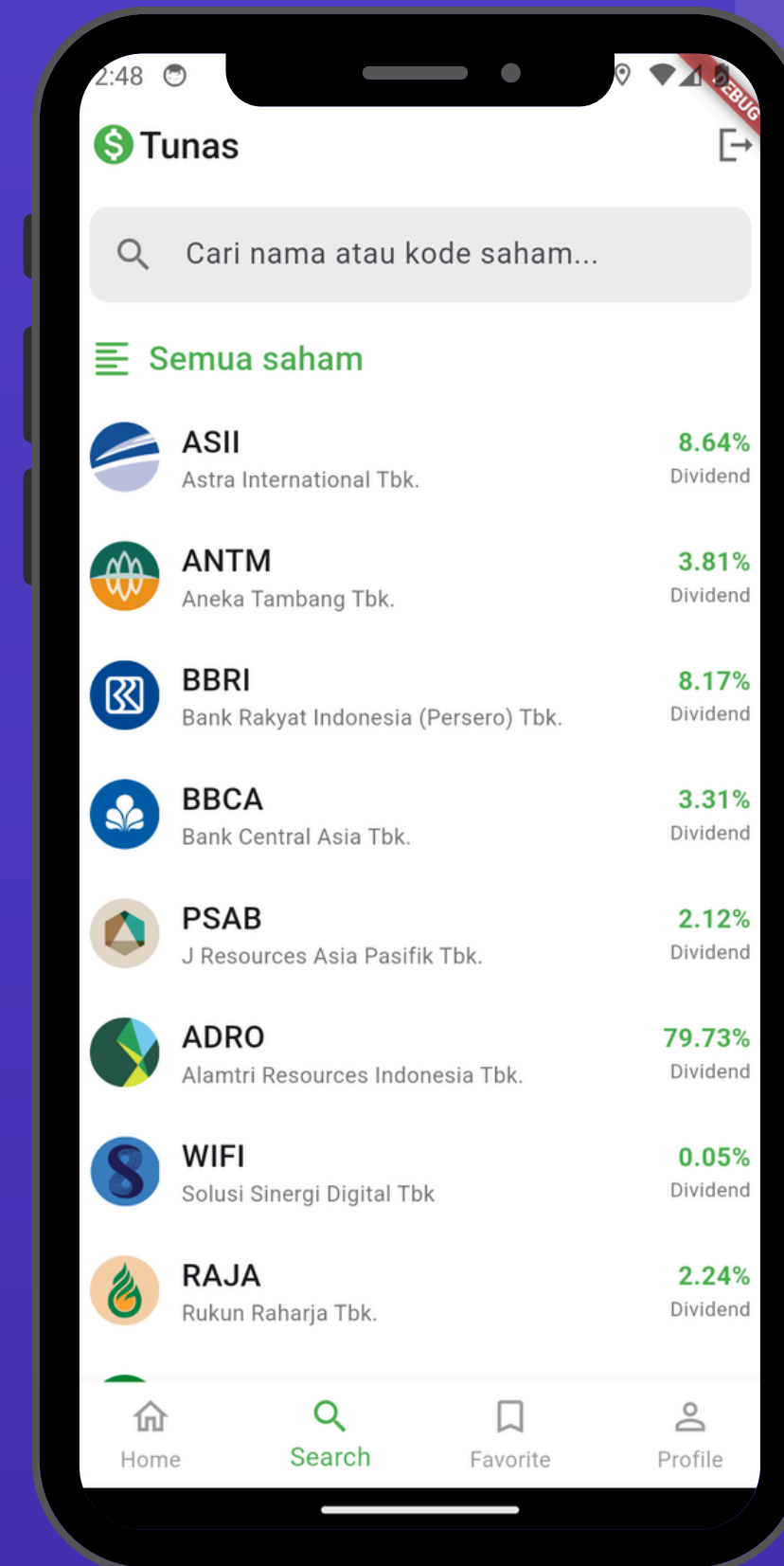
```
NotificationService().showNotification(  
  id: widget.stockData.id, // ID unik untuk setiap notifikasi  
  title: 'Favorit Ditambahkan',  
  body:  
    '${widget.stockData.name} (${widget.stockData.code}) berhasil ditambahkan ke daftar favorit Anda.',  
);
```



IMPLEMENTASI FITUR SEARCH

Fitur Search Saham

```
void _filterStocks() {  
    if (_searchQuery.isEmpty) {  
        _filteredStocks = _allStocks;  
    } else {  
        final queryLower = _searchQuery.toLowerCase();  
        _filteredStocks = _allStocks.where((stock) {  
            final nameLower = stock.name.toLowerCase();  
            final codeLower = stock.code.toLowerCase();  
            return nameLower.contains(queryLower) || codeLower.contains(queryLower);  
        }).toList();  
    }  
}
```



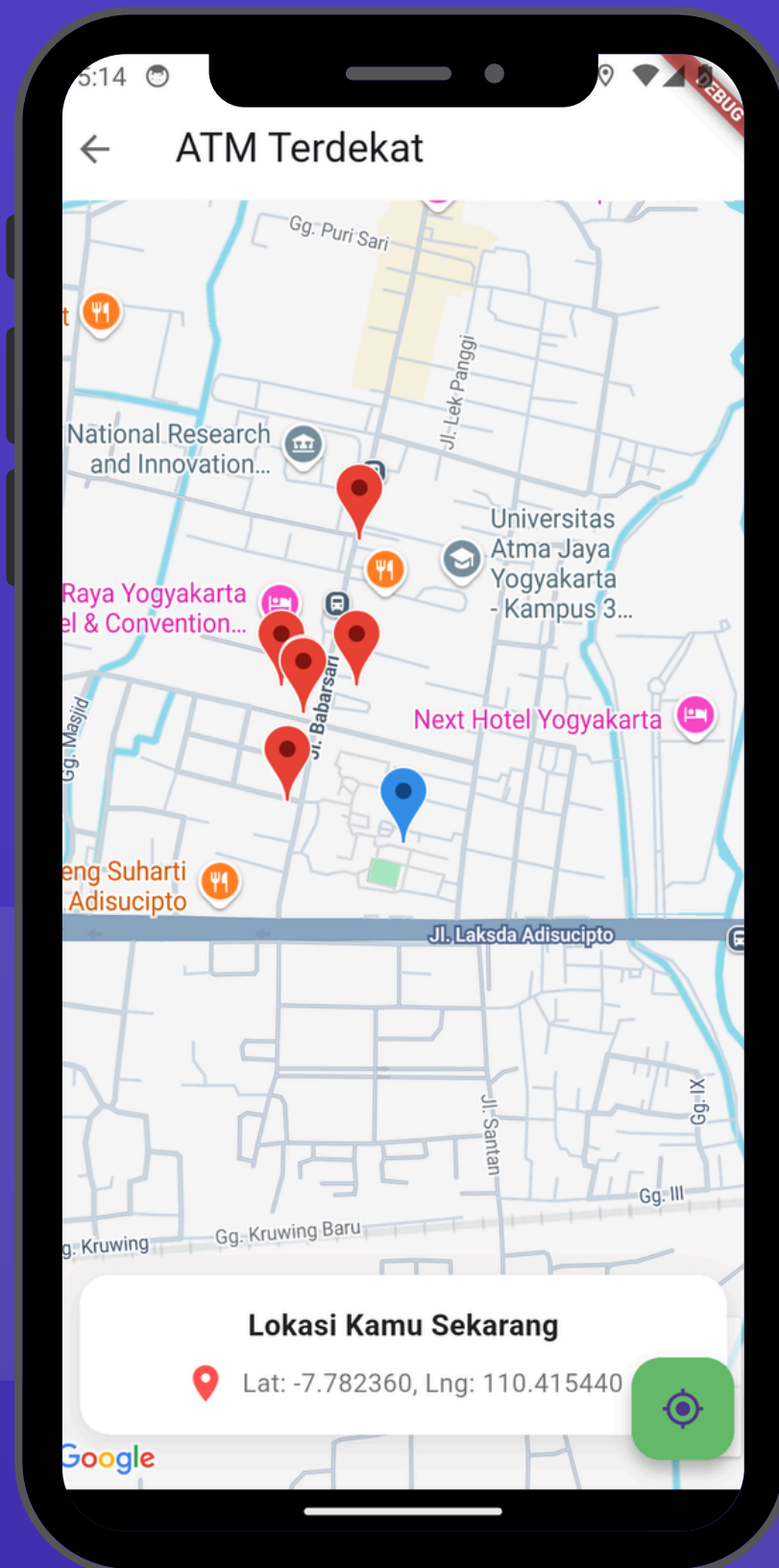
PROFILE DAN LOGOUT

Fitur Logout

```
actions: [  
  IconButton(  
    onPressed: () async {  
      await loginData.remove('username');  
  
      if (mounted) {  
        ScaffoldMessenger.of(context).showSnackBar(SnackBar(  
          content: Text("Berhasil logout"),  
          backgroundColor: Colors.green)); // SnackBar  
        Navigator.of(context)  
          .pushNamedAndRemoveUntil('/loginpage', (route) => false);  
      }  
    },  
    icon: Icon(Icons.logout_sharp),  
  ), // IconButton  
],
```



IMPLEMENTASI FITUR LBS



Implementasi LBS dengan API Google Maps

```
void startTracking() {
    Geolocator.getPositionStream(
        locationSettings: const LocationSettings(
            accuracy: LocationAccuracy.high,
            distanceFilter: 5,
        ),
    ).listen((position) {
        final newPosition = LatLng(position.latitude, position.longitude);
        if (!_hasFetchedATMs && mounted) {
            _fetchNearbyATMs(newPosition);
            setState(() {
                _hasFetchedATMs = true;
            });
        }
        if (mounted) {
            setState(() {
                _currentPosition = newPosition;
                latText = position.latitude.toStringAsFixed(6);
                longText = position.longitude.toStringAsFixed(6);
                _userMarker = Marker(
                    markerId: const MarkerId('user'),
                    position: newPosition,
                    infoWindow: const InfoWindow(title: "Kamu di sini"),
                    icon: BitmapDescriptor.defaultMarkerWithHue(
                        BitmapDescriptor.hueAzure),
                ); // Marker
            });
        }
        _mapController?.animateCamera(CameraUpdate.newLatLng(newPosition));
    });
}
```




TERIMA KASIH

Nandisya Faiz